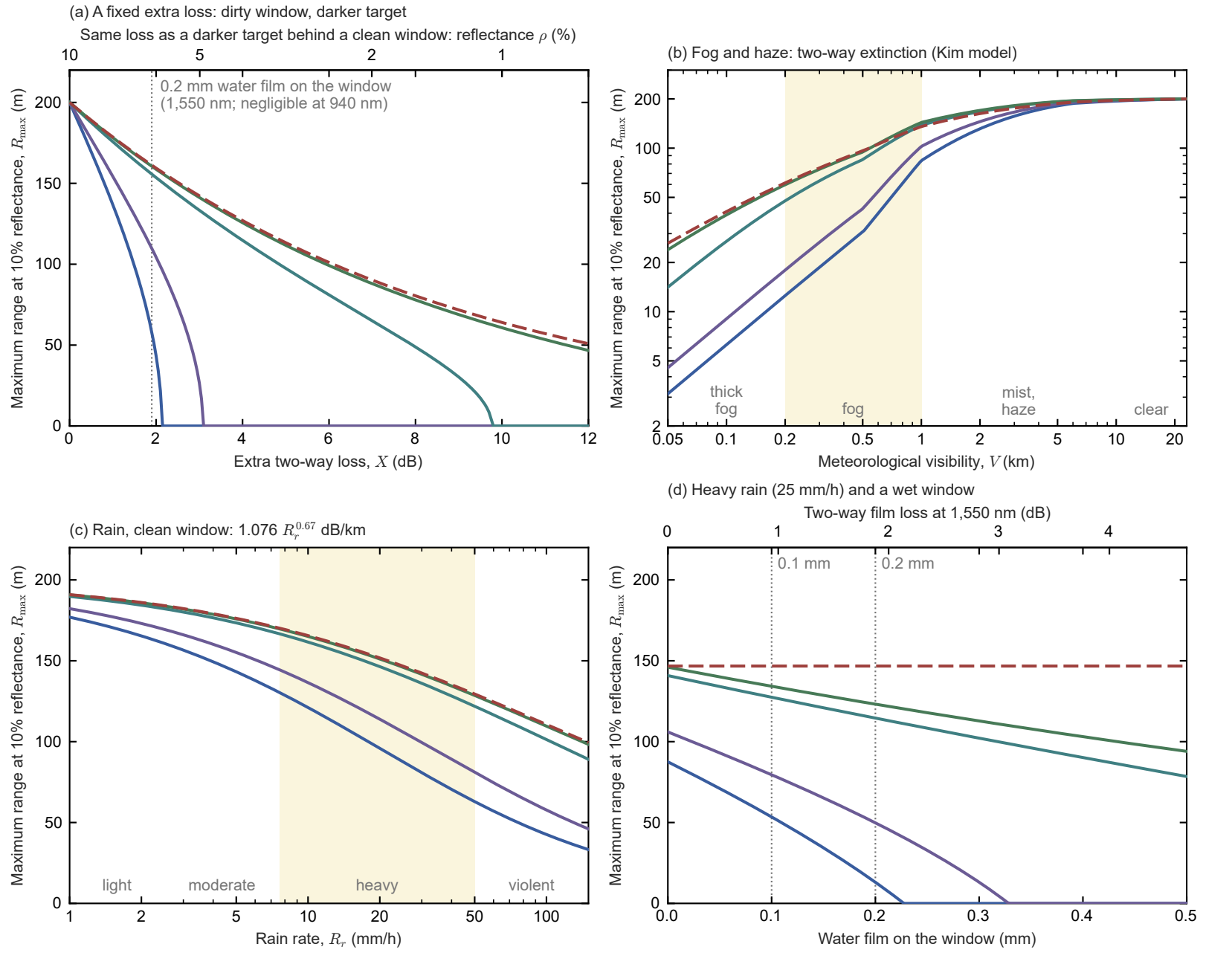
(a) A fixed extra loss: dirty window, darker target
Same loss as a darker target behind a clean window: reflectance ρ (%)
10
5
2
1
0.2 mm water film on the window
(1,550 nm; negligible at 940 nm)
Maximum range at 10% reflectance, R_{max} (m)
Extra two-way loss, X (dB)
(b) Fog and haze: two-way extinction (Kim model)
thick fog
fog
mist, haze
clear
Meteorological visibility, V (km)
(c) Rain, clean window: 1.076 $R_r^{0.67}$ dB/km
light
moderate
heavy
violent
Rain rate, R_r (mm/h)
(d) Heavy rain (25 mm/h) and a wet window
Two-way film loss at 1,550 nm (dB)
0.1 mm
0.2 mm
Water film on the window (mm)
Each design's energy per point is set so that it reaches 200 m on a 10% target in clear air
SPAD ToF, 940 nm, $D = 25$ mm lens: 74 nJ per point
FMCW, 1,550 nm, $D = 25$ mm: 312 nJ per point
FMCW, $D = 22$ mm (best for 200 m): 303 nJ per point
FMCW, $D = 13$ mm: 489 nJ per point
FMCW, $D = 7$ mm: 1,522 nJ per point